বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি (বাইউস্ট), কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Fall 2025

Department of Computer Science and Engineering
Level- 2, Term-II
Course Code: MATH 221
Course Title: MATH-IV
Credit Hour: 03

Notes:

a. Figure on the right of each question indicate marks for respective question. **Time : 1 hr. 30 min.**
b. **Answer any 03 (Three) of the following 04 (Four) questions including Q1.** **Full Marks : 90**

1. a. **Write** down two advantages of $A.M, G.M, H.M$ and show that $(a+b+c)(ab+bc+ca) \geq 9abc$. 10

   b. Suppose a survey about 50 students of BAIUST has done and the following data represents their Midterm marks (Math) 10

   85, 54,72, 73, 74, 82, 36, 45, 42, 66, 55, 72, 86, 51, 69, 77, 91, 62, 53, 51, 79, 38, 53, 60, 74, 58, 65, 72, 73, 56, 42, 85, 54,72, 73, 56, 42, 85, 54,82, 86 72, 48, 37, 56, 67, 73, 74, 82, 72.

   Construct a frequency distribution table using the given observations, compute the $G.M$ and find a conclusion for the survey by using Pie-chart (angle).

   c. Using frequency distribution table 1(b) **find** $Q3$, $D4$ and $P25$ from the data set. 10

2. a. Three cards are drawn from in succession, without replacement, from an ordinary deck of playing cards. **Find** the probability that the first card is a red ace; the second card is a 10 or a jack and the third card is greater than 3 but less than 7. 08

   b. A disease is known to be present in 2% of a population. A test is developed to help determine whether or not someone has the disease. 14
   Given that a person has the disease, the test is positive with probability 0.95.
   Also given that a person does not have the disease, the test is positive with probability 0.03.
   i) **Draw** a tree diagram to represent this information
   A person in selected at random from the population and tested for this disease.
   ii) **Compute** the probability that the test is positive.

   A doctor randomly selects a person from the population and tests him for the disease.
   Given that the test is positive,
   iii) **Identify** the probability that he does not have the disease.
   iv) **Discuss** on the usefulness of this test.

c. Given the following joint density function of X and Y: 08

$$f(x,y)=\begin{cases}4xy, & 0\le x\le 1, 0\le y\le 1\\ 0, & \text{elsewhere}\end{cases}$$

Show that X and Y are independent.

3. a. A fair is tossed three times. **Compute** the probability that:(a) exactly two runs and (b) Less than two runs. 08

b. Three coins are tossed, **Show** that the events " heads on the first coin" and the event " tails on the last two" are independent. 07

c. Examination results of 150 students showed that 95 students passed mathematics, 75 students passed economics and 135 students passed at least one of the above subjects. A student is selected random. **Find** the probability that the student 15

(a) Passed both mathematics and economics.

(b) Failed both the subjects.

(c) Passed mathematics but failed economics.

4. a. Consider the following joint density function of X and Y 15

$$f(x,y)=\frac{x+y}{8},\quad 0\le x\le 2, 0\le y\le 2$$

(i) **Show** that it is a joint density function.

(ii) **Compute** the marginal distributions of X and Y.

(iii) And conditional distributions of X and Y.

b. A department store has 6 television sets of which 2 are defective. A purchaser makes a random choice of 3 sets. If X is the number of defective sets purchased, **find** the probability distribution of X 08

c. There are 5 people in a room. **Find** the probability that at least two of them will have the same birthday. 07

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Department of Computer Science & Engineering
Level- II Term-II

Midterm Examination, FALL 2025
Course Code: CSE 223, Credit Hour-03
Course Title: Digital Logic Design
Marks: 90, Time: 1 hour 30 minutes

Notes:

a. Each question carries 30 marks.
b. Figure on the right of each question indicate marks for respective question.
c. **Scientific Calculator is not allowed.**
d. Required Postulates and Theorems of Boolean algebra are given in Table 1.

Course Outcomes:

[CLO-1]: Remember and understand the number system and Boolean algebra and basic properties of Boolean algebra to simplify simple Boolean functions.
[CLO-2]: Understanding and applying the tabulation and Karnaugh map methods for simplifying combinational circuits.

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1. a. Compare one's complement and two's complement representation methods. [CLO-1] — 10

   b. Obtain the weighted binary code for the decimal digits 7, 8, and 9 using the following weightings: [CLO-1] — 2x5=10

      I. The **84-2-1** code.
      II. The **85421** code.

OR

   b. Prove the following identity using Boolean algebra laws: [CLO-1] — 10

$$A.B + C.D = (A+C)\ (A+D)\ (B+C)\ (B+D)$$

   c. Simplify the following Boolean function to a minimum number of literals. [CLO-1] — 10

$$xy+x'z + yz$$

2. a. **Generate** the Gray code sequence for binary numbers 000 to 111. [CLO-1] — 10

b. Convert the following numbers as directed: [CLO-1] 5x2=10

I. **0.625** (decimal) to binary

II. decimal $(37)_{10}$ to BCD and coding.

OR

b. Find the product of sum (POS) and Sum of Product (SOP) for the following function, [CLO-1] 10

$$F(A, B, C) = B'C$$

c. What is Venn diagram? Illustrate the Absorption law, $X+XY = X$ by using Venn diagram. [CLO-1] 10

OR

c. Solve the following function using minimum no of gates/literals. [CLO-1] 10

$$ABCD+ABCD' + A'BCD + A'BCD'$$

3. a. Describe what happens when all cells in a K-map are 1's. Using K-map, obtain the simplified expressions in **product of sums** for the following Boolean functions: [CLO-2] 15

$$F(A, B, C, D, E) = \pi(0, 14, 16, 17, 21, 25, 29)$$

b. Explain how don't care conditions provide flexibility in K-map simplification. Simplify the following function using K-Map. [CLO-2] 15

$$F = A'B'C'+B'CD'+A'BCD'+AB'C'$$

Table 1:

1	Postulate 2	(a) X+0 = X	(b) X.1 = X
2	Postulate 5	(a) X+X′ = 1	(b) X. X′ = 0
3	Theorem 1	(a) X+X = X	(b) X.X = X
4	Theorem 2	(a) X+1 = 1	(b) X.0 = 0
5	Postulate 3, Commutative	(a) X+Y = Y+X	(b) XY = YX
6	Theorem 4, Associative	(a) X+(Y+Z) = (X+Y)+Z	(b) X(YZ) = (XY)Z
7	Postulate 4, Distributive	(a) X(Y+Z) = XY+XZ	(b) X+YZ = (X+Y) (X+Z)
8	Theorem 5, De-Morgan	(a) $(X + Y)' = X' Y'$	(b) $(XY)' = X'+Y'$
9	Theorem 6, Absorption	(a) X+XY = X	(b) X(X+Y) = X

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Department of Computer Science and Engineering

Mid-term Examination, Fall 2025
Level-2, Term-II
Course Code: GED 221
Course Title: Engineering Economics

Notes: **Time: 1.5 hours**
Full Marks: 90

a. Answer all of the following *03* (*Three*) questions.
b. Each set of questions carries 30 marks.
c. Figure on the right of each question indicate marks for respective question.
d. The course learning outcomes are:
   i. Develop preliminary knowledge of economic concepts for understanding of the national and international economy.
   ii. Understand how markets work, when markets fail and how the government can affect market outcomes
   iii. Evaluate macroeconomic policies based on their effects on the economic variables of interest.
   iv. Apply different techniques of investment decisions.

1.	a)	Define utility. Differentiate cardinal utility from ordinal utility.		[15]
	b)	Explain the law of Equi-marginal utility with the help of numerical example.	[CLO1]	[15]
2.	a)	Define demand and explain the factors that affect it.		[15]
	b)	Differentiate graphically between a change in demand and a change in quantity demanded.	[CLO2]	[15]
OR	a)	Distinguish individual supply from market supply with the help of diagrams.		[15]
	b)	Define Market equilibrium. Given the demand function Q_d=100−2P and the supply function Q_s=20+3P, determine the equilibrium price and quantity. Plot the values in figure.	[CLO2]	[15]
3.	a)	Illustrate the concept of an indifference curve using a diagram.		[15]
	b)	Examine the main properties of an indifference curve with the help of diagrams.	[CLO1]	[15]



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Mid Term Examination, Fall 2025
Department of Computer Science and Engineering
Level-2 Term-II
Course Code: CSE-221
Course Title: Computer Algorithms and Complexity
Credit Hour: 03
Exam Duration: 1.5 Hours
Full Marks: 90

Notes:

a. The figure on the right of each question indicates marks for each question.
b. Answer ***all THREE*** questions.
c. Course Learning outcomes are:

CLO1: Analyze the asymptotic performance and write rigorous correctness proofs for algorithms.
CLO2: Estimate the performance of various algorithms and data structures.
CLO3: Apply important algorithmic design paradigms and methods of analysis for problem solving.
CLO4: Device efficient algorithms in common engineering design situations.

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1. a) Why does Selection Sort perform fewer swaps than Bubble Sort, and how does this affect its efficiency in environments where swapping is costly? **[CLO2 → C4]** **10**

OR,

Analyze the time complexity of the Tower of Hanoi problem for n disks. Derive the recurrence relation and solve it to get the number of moves needed. **[CLO2 → C3]**

b) Perform heap sort on the dataset 65,24,38,41,12,30,5565,24,38,41,12,30,55, and describe the construction of the max-heap and the sorting process step by step. **[CLO2 → C3]** **10**

c) How can the singly-even algorithm be systematically applied to construct a **6×6** magic square, and why does this method ensure that every row, column, and diagonal add up to the magic constant of 111, as derived from the formula $M_n = \frac{n(n^2+1)}{2}$? **[CLO2 → C3]** **10**

2. a) Given the array A=92, 3, 57, 41, 68, 12, 85, 29, 7 apply Merge Sort step by step. For each step, clearly show the recursive division into subarrays, the merging process at each level, and the final sorted array. **[CLO3 → C3]** **10**

b) Write pseudocode for binary search and trace the algorithm for the array 5, 15, 25, 35, 45, 55, 65 with target value 40. Show all intermediate steps and explain why the search fails. **[CLO3 → C3]** **10**

OR,

Write pseudocode for inserting a value into a Binary Search Tree (BST). Ensure that the pseudocode maintains the BST property: the left subtree contains values smaller than the node, and the right subtree contains values greater than the node. **[CLO3 → C3]**

c) Given the array A=76, 14, 53, 29, 87, 6, 42, 98, 21 apply Quick Sort step by step. For each step, clearly show the choice of pivot, the partitioning into subarrays, recursive sorting, and the final sorted array. **[CLO3 → C3]** **10**

3. a) You are given a knapsack with a maximum weight capacity of **50 units**. The following table lists the available items, each with a specific weight and value: **10**

Item	Weight (units)	Value (units)
A	8	64
B	15	90
C	27	135
D	13	91
E	20	100
F	4	28
G	10	50
H	18	81

Apply the Greedy Knapsack algorithm to determine the maximum value that can be carried in the knapsack. **[CLO3→C3]**

OR,

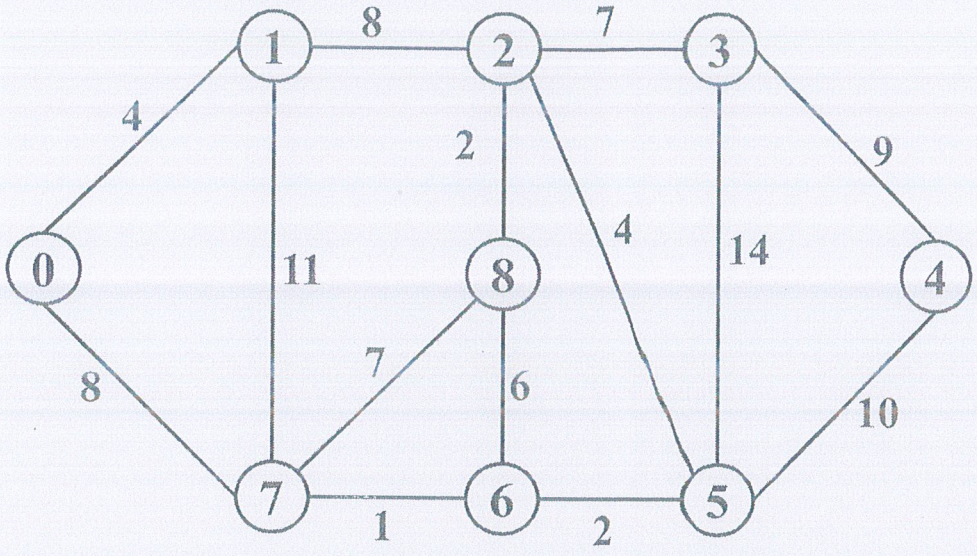
You are given the following set of jobs, each with a deadline and a profit. Find the maximum profit using the Job Sequencing with the Deadline algorithm.

Job	Profit	Deadline
J1	35	3
J2	30	4
J3	25	4
J4	20	2
J5	15	3
J6	12	1
J7	5	2

Given the following set of jobs, each with a deadline and a profit, find the maximum profit using the Job Sequencing with Deadline algorithm. **[CLO3→C3]**

b) Apply Dijkstra's algorithm to find the shortest path from node P to node S in a weighted graph with the following edge weights: P→Q, 6; P→S, 2; Q→R, 3; Q→S, 4; R→S, 2; P→R, 5. Show the sequence of nodes visited and the final shortest path distance. **[CLO3→C3]** **10**

c) Determine the Minimum Spanning Tree (MST) of the given weighted graph using both Kruskal's and Prim's algorithms, showing all necessary steps in the construction. **[CLO3→C3]** **10**



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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Fall 2025

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE-225

Course Title: Numerical Methods

Credit Hour: 03

Exam Duration: 1 hour 30 Minutes

Full Marks: 90

Notes:

a. Figures on the right of each question indicate the marks for the respective question.
b. Answer all **THREE** questions.
c. The Course Outcomes are:

CO1: Recall the basics of accuracy and finite precision, round-off errors, truncation errors, and stability of numerical algorithms.

CO2: Apply appropriate numerical techniques and calculate the computational cost in the fields of science and engineering.

CO3: Formulate numerical techniques for root finding, curve fitting, differentiation, and integration.

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1. a. How is accuracy affected during the process of numerical computing? Find out the precision and accuracy for the following numbers **10**

i) 80.500 ii) 0.0080775 iii) 9000 iv) 05.00023410 v) 55 **[CO1→C3]**

b. Add and subtract the numbers 0.735916 E4 and 0.635742 E6 using computer arithmetic algorithm. **[CO1→C3]** **10**

c. Write an algorithm for non-decimal to decimal conversion system. Convert $(1101.1101)_2$ to decimal. **[CO1→C3]** **10**

OR

a. A computer represents numbers in floating-point form with a 6-digit mantissa and 2-digit exponent. **10**

(a) Find the range and precision of numbers that can be represented.

(b) Compute the result of adding 1.23456×10^2 and 9.87654×10^1 using proper normalization.

[CO1→C3]

b. If the true value of π is 3.1415926 and the approximate value is 3.14, calculate the absolute error, relative error, and percentage error. **[CO1→C3]** **10**

c. What is chopping? When does it occur? Explain with an appropriate example. **10**

[CO1→C4]

2. a. Apply the Bisection method to find the real root of $x^3 - 4x - 9 = 0$ in the interval [2,3]. Compute the root accurate to three decimal places. Show all iterations and explain how the bracketing interval changes at each step. **[CO2, CO3→C3]** **15**

   b. Explain how the choice of initial guess affects the convergence of the Fixed Point and Newton-Raphson methods. Provide an example where poor initial guesses lead to divergence. **[CO2, CO3→C4]** **10**

   c. Calculate the approximate value of $\Delta^2(\cos 3x)$ where $h = 1$. **[CO2, CO3→C3]** **5**

3. a. Use the Regula Falsi method to find the root of $f(x) = 3x - cosx - 1$ correct it to three decimal places. Show all intermediate calculations. **[CO2, CO3→C3]** **15**

   b. Utilize the Newton-Raphson method to find the root of $x^3 - x - 1 = 0$ correct up to three decimal places assuming $x_0 = 1$. **[CO2, CO3→C3]** **15**